Disaster Survival Simulator

A Scenario-Based Flood Preparedness and Response Learning Game

Course	Disaster Management (CE8.401)
Project Scope	First six flood games: Go-Bag, Build the Raft, Flood Rescue, Field Triage, Sinking Car, SOS Signaling
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Repository	`https://github.com/Sahil4804/Disaster_Awareness_Game`
Branch / Commit Reviewed	`old-state / 0916f5c`
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Project identity: Instead of limiting the work to a planned concept or a comic-style question-answer activity, the project implements playable, pressure-based disaster scenarios. Players make choices, experience consequences, receive feedback, and see their mistakes summarized in a readiness dashboard.

1 Abstract

Flood education is often delivered through posters, short videos, comics, or quiz questions. Such formats can introduce important terms, but they do not always train learners to act under time pressure, prioritize resources, or reflect on mistakes. Our project addresses this gap through a browser-based Disaster Survival Simulator built in React. The current report focuses on the first six flood games, which form one continuous story: a flash flood warning arrives, the player packs an evacuation go-bag, builds an improvised raft from garage materials, rescues stranded neighbors, triages and treats survivors at a relief shelter, escapes a sinking vehicle, and signals rescuers. The implementation combines side-scrolling exploration, top-down raft navigation, medical sorting, resource collection, quick-time escape tasks, condition-based SOS signaling, scoring, pass/fail feedback, and a readiness dashboard. The result is not only a game plan, but a playable scenario-based learning system. It helps users understand why a decision is safe or unsafe, where they lost points, and what they should improve before replaying. The dashboard converts attempts into learning evidence by showing readiness level, module progress, skill gaps, strengths, and next steps. This makes the project suitable for student learning, public awareness demonstrations, and disaster-preparedness training discussions.

2 Introduction

2.1 Background and Motivation

Floods are among the most common disaster events and can escalate quickly from warning to evacuation, road blockage, vehicle submergence, medical stress, and rescue communication. Preparedness guidance often explains what to pack, how to avoid floodwater, when to seek shelter, and how to signal for help. However, learners may still struggle when these decisions are placed in sequence. They may know that a torch is useful, but forget it while packing under time pressure. They may know that a raft requires buoyancy, but choose weak material under stress. They may know SOS exists, but not which signal works in fog, rain, sunlight, or night.

The motivation of this project is to make disaster education experiential. The game turns awareness content into a sequence of playable decisions so students can practice the logic of survival before a real event. The experience is designed around mistakes as learning moments: every module gives feedback, records score data, and supports replay.

2.2 Problem Statement

Traditional disaster learning games are frequently story-based, comic-based, or quiz-based. These formats ask users to choose answers, but the interaction often ends after the answer is marked correct or wrong. Our project asks a broader question:

> How can a flood-awareness game teach practical preparedness and response skills through scenario-based gameplay, measurable decisions, and dashboard feedback?

2.3 Scope

The repository contains a larger Disaster Survival Simulator interface with more planned flood modules and locked disaster types. This report considers only the first six implemented flood games because they form the complete story requested for evaluation:

1. Flood is about to come, so the player prepares a go-bag.
2. The player collects resources in a garage and builds a raft.
3. The raft is used to rescue people and bring survivors to safety.

4. The player reaches relief and medical sources, collects medicine, and treats people.
5. The player follows routes to escape, but the car sinks and must be escaped.
6. After escaping, the player performs SOS signaling to attract rescue.

2.4 Target Users and Objectives

The target users are school or college students, community volunteers, and general citizens learning basic flood preparedness. The learning objectives are:

- identify essential emergency supplies and avoid low-value or unsafe items;
- understand improvised raft material selection and buoyancy tradeoffs;
- practice rescue navigation while avoiding electrical and debris hazards;
- triage survivors and match medical needs with appropriate treatment items;
- follow safe behavior during route selection and sinking vehicle escape;
- choose suitable SOS signals for different visibility and weather conditions;
- use dashboard feedback to understand mistakes and improve across attempts.

3 Disaster Context and Domain

The first six games focus on flood preparedness and response. The domain content draws from standard flood-safety ideas: keeping an emergency kit, avoiding moving floodwater and submerged roads, evacuating when necessary, reducing exposure to contaminated floodwater, using safe rescue communication, and treating basic post-flood medical risks. The simulator does not claim to replace official training. Instead, it converts the logic of official guidance into interactive learning.

Flood Risk	Learning Need	Game Translation
Sudden warning and evacuation	Pack essentials quickly and avoid useless weight	Timed go-bag selection with weight and essential-item scoring
Flooded streets and lost mobility	Understand makeshift flotation and material reliability	Garage collection and raft stress test
Stranded people	Rescue without increasing danger	Night raft navigation, victim pickup, shelter drop-off, hazard avoidance
Injured or ill survivors	Prioritize by severity and treat with correct supplies	Triage ordering, supply run, and treatment matching
Submerged car	Escape before cabin flooding becomes fatal	Route choice followed by staged sinking-car escape actions
Need for external rescue	Signal according to conditions	SOS rounds using mirror, flare, whistle, flag, and fire

Table 1: Flood-domain risks converted into playable learning tasks.

4 Methodology / Approach

4.1 Design Approach

The project uses a scenario-based design rather than an isolated quiz format. The six modules are connected by one flood story. Each module has a clear pressure condition, such as a countdown, weight limit, limited visibility, boat capacity, rising water level, or weather-dependent signaling. This creates active practice instead of passive reading.

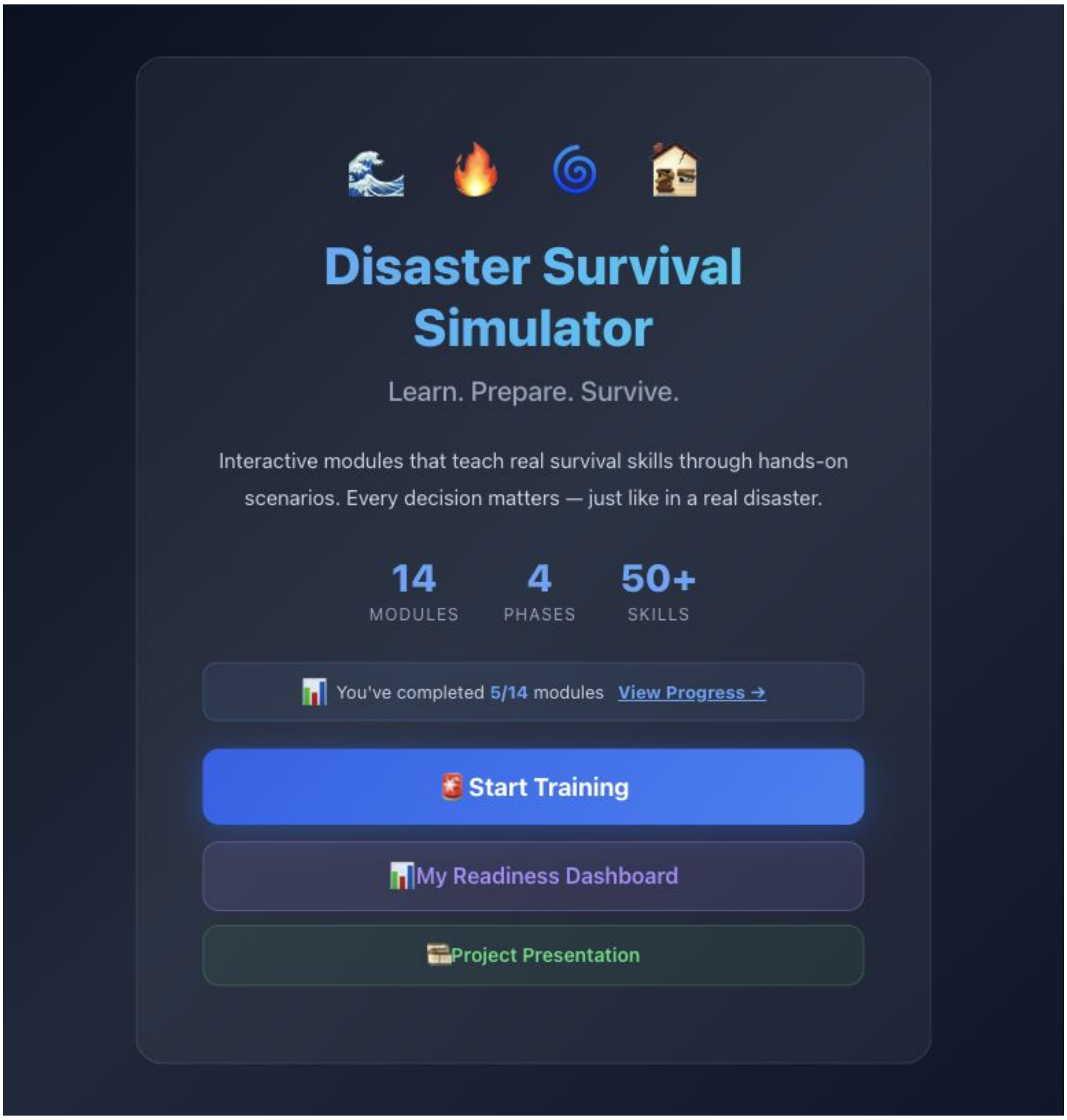

Figure 1: Main menu of the Disaster Survival Simulator, showing the project identity and entry points for training, dashboard review, and presentation.

Phase	Scenario Event	Player Work
Preparedness	Flood warning arrives	Pack a practical go-bag
Improvisation	Garage becomes the resource area	Build a working raft
Rescue	Neighborhood is flooded at night	Navigate, identify victims, and deliver survivors
Medical response	Survivors reach shelter	Triage, collect supplies, and treat cases
Escape	Evacuation route fails and car sinks	Escape the vehicle through ordered actions
Communication	Player is outside but still needs rescue	Send the right SOS signal

Table 2: Story arc used by the first six flood games.

4.2 Technology Stack

The project is implemented as a Vite and React application. Important files reviewed include the app router, shared game context, the six flood module components, the module selector, the metrics dashboard, and the sources panel. The application uses browser local storage to preserve scores, completed modules, attempts, best score, and dashboard progress.

4.3 Implementation Pattern

Each module follows a similar structure:

- a self-contained React component controls the gameplay state;
- timed or spatial interaction creates pressure;
- the module computes a numeric score and pass/fail result;
- feedback explains the consequence of player choices;
- the result is recorded through the shared game context;
- the dashboard turns module attempts into readiness insights.

4.4 First Six Game Modules

Module	File	Gameplay	Learning Outcome
Go-Bag Challenge	`Module1_GoBag.jsx`	Side-scrolling home exploration. The player collects emergency items under time and weight limits.	Learns essential kit selection: water, medical supplies, torch, documents, rain protection, and avoiding low-value items.
Build the Raft	`Module2_HomeDefense.jsx`	Garage collection and progressive raft assembly with hull, binding, deck, and propulsion choices.	Learns buoyancy, binding strength, platform stability, and paddling control.
Flood Rescue	`Module3_YardLockdown.jsx`	Top-down night raft navigation, survivor pickup, debris clearing, shelter drop-off, and electrical hazard avoidance.	Learns safe rescue routing, victim identification, limited capacity, and hazard awareness.
Field Triage	`Module4_FieldTriage.jsx`	Triage ordering, raft-based supply run to medical sources, and treatment selection.	Learns medical prioritization, supply matching, and correct treatment for dehydration, diabetes, hypothermia, fractures, and cuts.
Sinking Car	`Module5_SinkingCar.jsx`	Route decision followed by escape actions: calm down, unbuckle, open or break window, and swim out.	Learns that flooded roads are dangerous and that escape requires ordered action before water level rises.

Module	File	Gameplay	Learning Outcome
SOS Signaling	`Module6_SOSSignaling.jsx`	Weather and visibility rounds where the player selects and performs the right signaling method.	Learns that mirror, flare, whistle, flag, and fire are condition-dependent rescue signals.

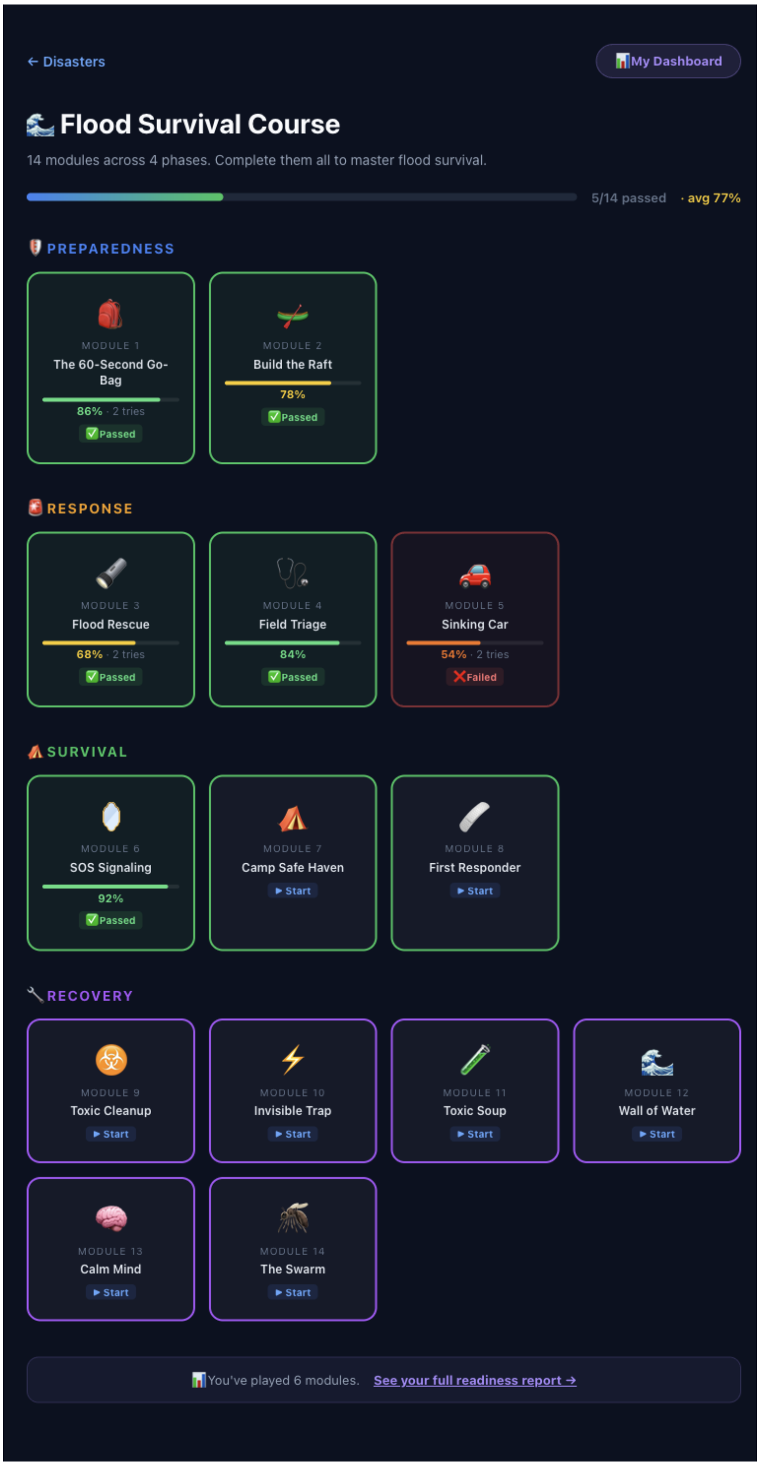

Figure 2: Flood module-selection screen with progress, score bars, pass/fail labels, attempts, and phase grouping across the scenario sequence.

5 Data Description

This project does not use a survey dataset or machine-learning dataset. The main data are gameplay states, code-defined item lists, victim profiles, resource locations, scoring rules, and local progress records. These are meaningful because they represent the educational model of the simulator.

Data Type	Where It Appears	Purpose
Item catalogues	Go-bag and raft modules	Define correct, weak, unsafe, or low-priority choices
Victim profiles	Rescue and triage modules	Represent survivor conditions and medical priority
Hazard zones	Rescue, triage, and route modules	Teach avoidance of electrical, road, and floodwater danger
Timers and pressure states	All six games	Simulate urgency and decision-making under stress
Scores and attempts	`GameContext.jsx` and browser local storage	Track progress, attempts, pass counts, and best score
Dashboard metadata	`MetricsDashboard.jsx`	Convert scores into readiness grade, skill analysis, strengths, gaps, and next steps
Source metadata	`moduleSources.js` and `SourcesPanel.jsx`	Connect learning content with official disaster guidance

Table 4: Project data types observed in the source files.

6 Results and Analysis

6.1 Implemented Outputs

The project implements a playable flood-learning path instead of stopping at a conceptual plan. The first six games together cover preparedness, response, treatment, escape, and rescue communication. Each game has a scoring rule connected to safety behavior:

- Module 1 rewards essential go-bag items and penalizes excess weight or traps.
- Module 2 rewards correct raft materials and blueprint reading.
- Module 3 rewards delivered survivors, avoiding shocks, and rescuing everyone in time.
- Module 4 rewards correct triage, supply collection, and treatment decisions.
- Module 5 measures escape success by remaining safety margin before flooding.
- Module 6 rewards correct signaling tools and performance in weather-specific rounds.

6.2 Dashboard Learning Result

The dashboard is one of the strongest educational parts of the project. It does not only show a final score. It converts gameplay into reflection by presenting:

- overall readiness grade and pass rate;
- active module count and completed module count;
- phase-wise preparedness, response, and survival analysis;

- module-by-module scores, attempts, and pass/fail status;
- skill categories such as priority, protocol, hazard, resource, speed, pressure, navigation, and engineering;
- strengths, weak areas, untested skills, and suggested next steps.

This means players can learn from mistakes after the game. For example, a low score in the go-bag module may show poor resource prioritization, while a low score in flood rescue may reveal navigation or hazard-awareness problems. The dashboard therefore turns the game into a feedback-driven learning system.

Figure 3: Readiness dashboard showing overall grade, phase-wise knowledge snapshot, module-by-module performance, failed attempts, retries, and skill-profile analysis.

6.3 Educational Value

The scenario chain makes the project memorable because one mistake can be understood within a story. Packing a poor bag affects preparedness. Building a bad raft affects mobility. Missing rescue hazards affects survivor safety. Wrong treatment affects medical outcomes. Delayed car escape affects survival. Wrong SOS signaling affects rescue chances. This is stronger than a single quiz because the learner practices the action, not only the answer.

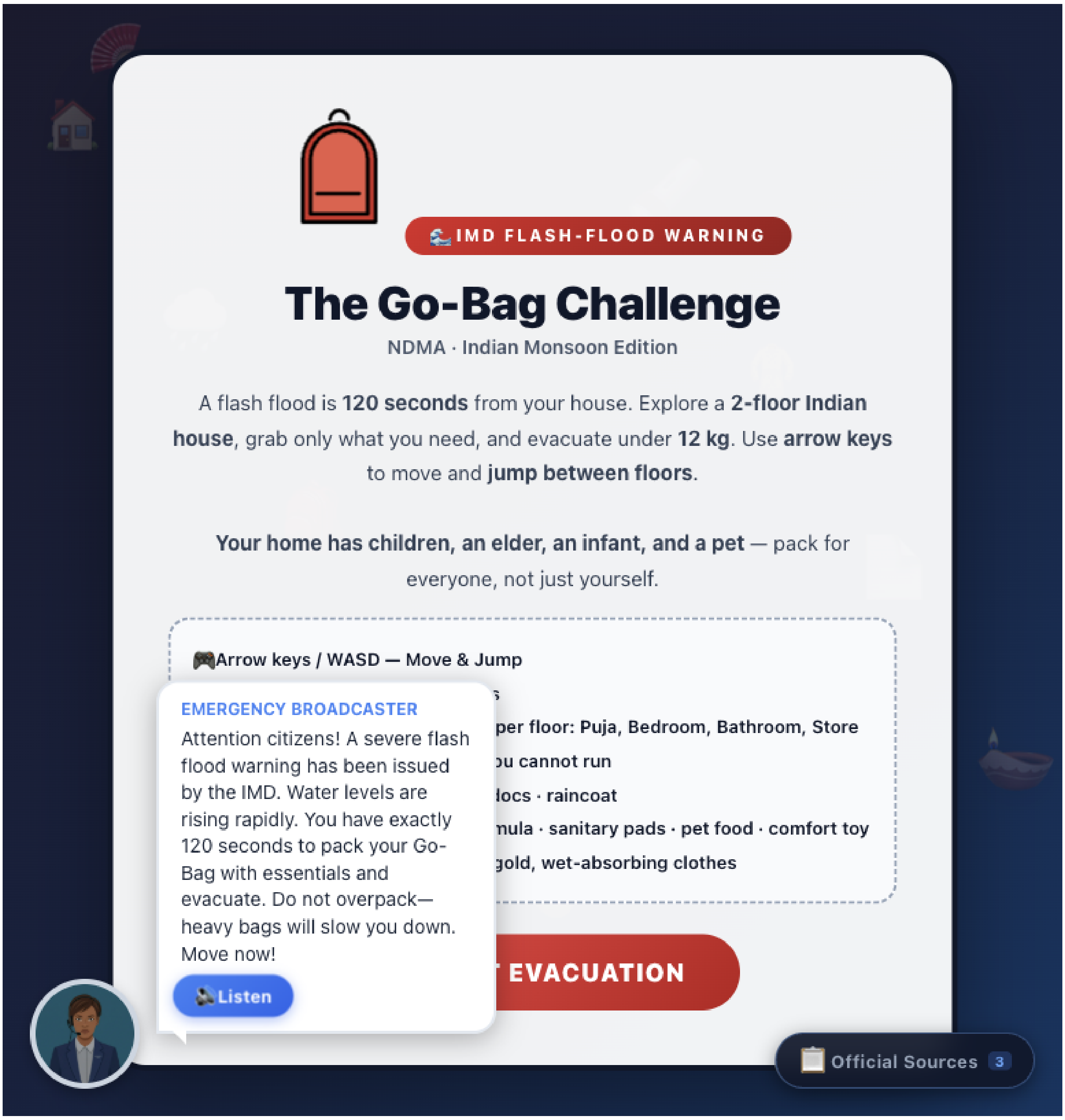

Figure 4: Module 1 visual: the Go-Bag Challenge begins with a flash-flood warning, time pressure, weight limit, family needs, and essential-item guidance.

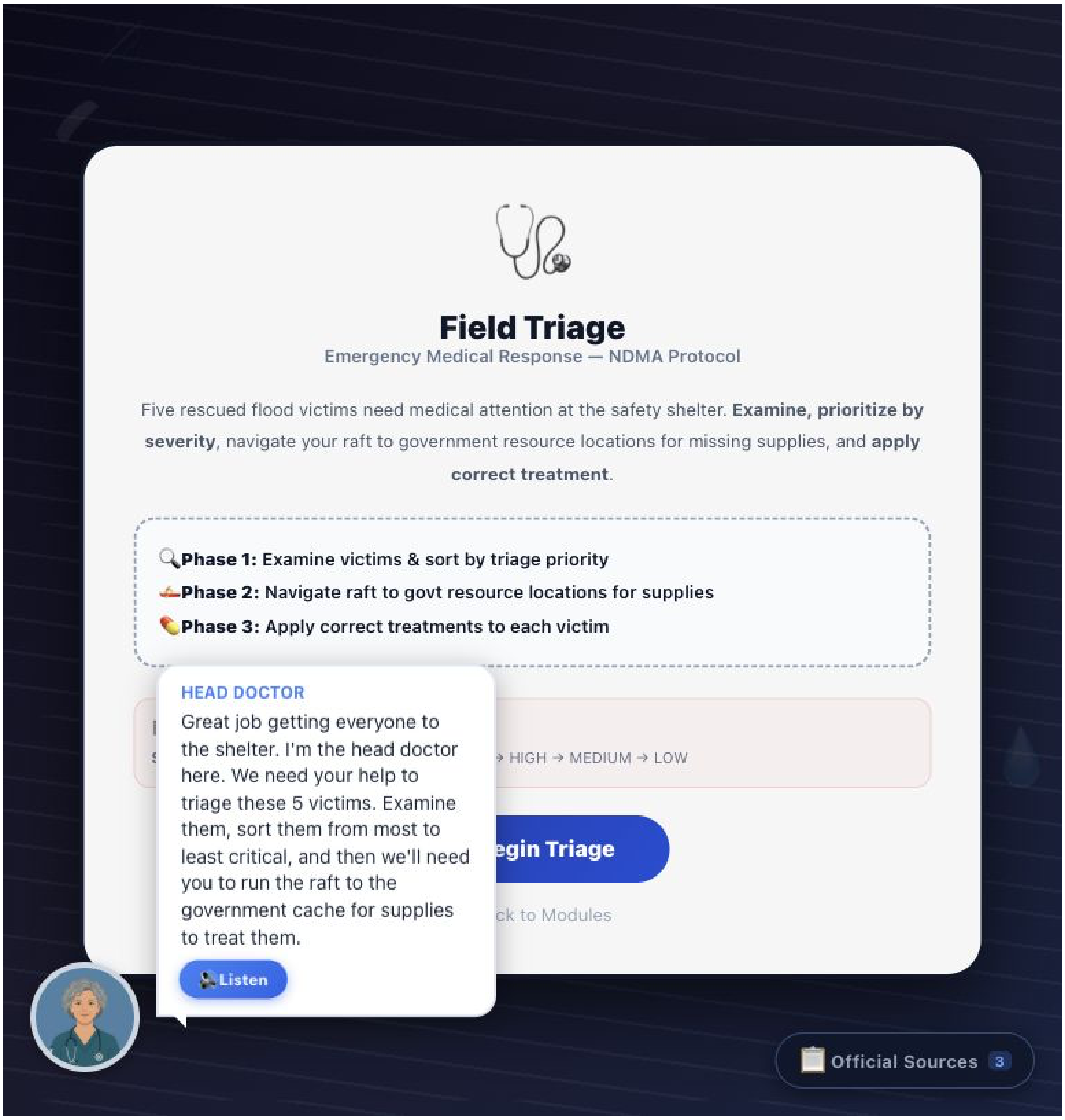

Figure 5: Module 4 visual: Field Triage introduces medical prioritization, supply collection, and correct treatment decisions after survivors reach shelter.

7 Comparison with Other Disaster Learning Games

Many simple disaster-awareness games are story-based, comic-style, or question-answer activities. They are useful for introducing concepts, but they often remain close to a classroom quiz. The user reads a situation, clicks an option, and receives a correct or incorrect response. Our project improves on that model by making the learner perform decisions inside an interactive flood sequence.

The main difference is that our project does not simply ask, "What should you do?" It asks the player to do it under simulated constraints. This makes the game more useful for learning because users can discover their own mistakes through play and then improve through the dashboard.

8 Limitations

Aspect	Typical Comic / Q&A Game	Our Disaster Survival Simulator
Learning mode	Reads story panels and answers questions	Performs tasks inside playable flood scenarios
Decision pressure	Usually low pressure	Timers, weight limits, visibility limits, capacity limits, and rising water
Skill coverage	Often focuses on recognition of facts	Covers packing, building, navigation, rescue, triage, treatment, escape, and signaling
Feedback	Correct or wrong answer feedback	Score, pass/fail, consequence text, retry, and dashboard analysis
Continuity	Isolated questions or short stories	Connected story from warning to SOS rescue
Reflection	Limited after-game analysis	Readiness dashboard with strengths, gaps, attempts, best scores, and next steps
Practical realism	Mostly conceptual	Scenario-based practice inspired by official safety guidance

Table 5: Comparison between conventional disaster-learning games and this project.

The project is functional and educationally strong, but it also has limitations:

- The report considers only the first six flood games, while the interface lists additional modules and locked disaster types.
- The simulation simplifies real rescue and medical decisions. It should support awareness, not replace professional training.
- The dashboard measures in-game performance, but a formal user study would be needed to measure real learning gains.
- Some source-panel metadata should be kept synchronized as module names and numbering evolve.
- The production build passes, but Vite reports a large bundle-size warning, so future optimization or code splitting would improve delivery.
- Accessibility improvements such as keyboard guidance, screen-reader labels, and adjustable difficulty would strengthen inclusivity.

9 Code Reproducibility

9.1 Reviewed Repository State

The reviewed project is the current branch `old-state` at commit `0916f5c`. The code was inspected after pulling the latest available changes from the tracked branch.

9.2 Important Source Files

File / Folder	Purpose
`src/App.jsx`	Main navigation, screen selection, and module routing.
`src/context/GameContext.jsx`	Shared game state, score recording, completed modules, and local-storage persistence.

File / Folder	Purpose
src/components/ ModuleSelect.jsx	Flood module list, progress summary, and entry points.
src/components/ MetricsDashboard.jsx	Readiness grade, phase stats, skill analysis, module insights, and next steps.
src/components/ SourcesPanel.jsx	Floating official-source panel used for learning support.
src/modules/flood/ Module1_GoBag.jsx	Go-bag preparation game.
src/modules/flood/ Module2_HomeDefense.jsx	Garage collection and raft-building game.
src/modules/flood/ Module3_YardLockdown.jsx	Flood rescue navigation game.
src/modules/flood/ Module4_FieldTriage.jsx	Triage, supply run, and treatment workflow.
src/modules/flood/ Module5_SinkingCar.jsx	Route decision and sinking vehicle escape.
src/modules/flood/ Module6_SOSSignaling.jsx	Condition-based SOS signaling rounds.
public/assets	Avatars, medical POV imagery, emoji, and icon assets.

9.3 Setup Instructions

```
git clone git@github.com-personal:Sahil4804/Disaster_Awareness_Game.git
cd Disaster_Awareness_Game
npm install
npm run dev
npm run build
```

9.4 Dependencies

Dependency	Role
react, react-dom	Component UI and rendering
vite	Development server and production build
@vitejs/plugin-react	React support in Vite
three, @react-three/fiber, @react-three/drei	Available for 3D modules in the broader project
gh-pages	Deployment helper

10 Conclusion

The Disaster Survival Simulator successfully demonstrates how disaster management education can move beyond passive awareness. The first six games form a coherent flood-survival journey: preparation, improvised mobility, rescue, medical response, vehicle escape, and rescue communication. Each game contains actionable mechanics, scoring rules, and feedback that connect learner choices to safety consequences. The dashboard strengthens the project by making mistakes visible and useful: players can see which modules and skills they lack, retry weak areas, and understand what to improve.

The main contribution is the shift from a comic or quiz-only design to a scenario-based simulator. The project does not simply ask learners what is correct; it places them inside a flood story and makes them practice the decision. This gives the system educational value for students, community awareness sessions, and disaster preparedness demonstrations.

11 References

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